

Homework 8

Due Friday, December 8

1. Consider a two-component complex scalar field $\Phi(\mathbf{x}, t)$,

$$\Phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}, \quad \Phi^\dagger = (\phi_1^*, \phi_2^*),$$

described by the (relativistic) Lagrangian density

$$\mathcal{L} = \eta^{\mu\nu} \partial_\mu \Phi^\dagger \partial_\nu \Phi - m^2 \Phi^\dagger \Phi,$$

where $\eta^{\mu\nu} = \text{diag}(1, -1, -1, -1)$, m^2 is a parameter, and summation over $\mu, \nu = t, x, y, z$ is implied. Multiplications of $\Phi^\dagger$ and Φ in $\mathcal{L}$ are row-by-column; for example, in the second term, $\Phi^\dagger \Phi = \phi_1^* \phi_1 + \phi_2^* \phi_2$. We use units in which $c = 1$.

(a) Considering all components of Φ and $\Phi^\dagger$ as independent variables, find the canonical momentum density corresponding to each.

(b) Verify that the Lagrangian density is invariant under transformations

$$\Phi(\mathbf{x}, t) \rightarrow U \Phi(\mathbf{x}, t),$$

where U is any coordinate-independent unitary 2×2 matrix with unit determinant. A general form of such a matrix is

$$U = \exp \left(\frac{i}{2} \sum_{a=1,2,3} \alpha_a \tau_a \right),$$

where τ_a are the Pauli matrices, and α_a are parameters; i is the imaginary unit. (Exponential of a matrix is defined by means of the Taylor series: $e^A = 1 + A + \frac{1}{2}A^2 + \dots$) Note that there are three independent parameters α_a . Find the corresponding three Noether charges, Q_a , $a = 1, 2, 3$.

(c) Express the Noether charges in terms of Φ , $\Phi^\dagger$, and the momentum densities found in part (a). Use these expressions to compute the Poisson bracket $\{Q_1, Q_2\}$.

2. (a) Consider a theory of a real scalar field $\psi(x, t)$ in one spatial dimension with a Lagrangian density that depends on the second spatial derivative:

$$\mathcal{L} = \mathcal{L}(\psi, \psi_t, \psi_x, \psi_{xx})$$

(subscripts denote partial derivatives). Applying Hamilton's variational principle (that is, searching for an extremum of $S = \int dt dx \mathcal{L}$ with suitable conditions at the boundary), derive the analog of the Euler-Lagrange equation for this case.

(b) Problem 10 to Chap. 13 of the textbook.